

CSC621 Data Structures and Algorithm

Lab Exercise 5 - SLL

This lab is based on your previous tutorials where an array was used to store marks, IDs and grades. In this exercise, you are required to use a linked list of students instead of an array.

Question 1 – Student attributes part of Node class

1. Download SLL program from Moodle.
2. Modify Node class to replace *data* attribute with student *name*, *marks* and *grade*.
3. Update the constructors, accessors and modifiers where needed.
4. Implement the following menus in your main program:

Choose an option:

I - Insert

D - Display

S – Stop

where **Insert** is used to insert a student detail to the beginning of the list, **Display** is to show all the data from the list in a tabular format and **Stop** is to terminate the program.

5. Test your program by inserting and displaying three student details.

Question 2 – Separate Student Class

Modify Question 2 with the following:

1. Download SLL program from Moodle.
2. Create *Student* class that encapsulates *name*, *marks* and *grade*.
3. Create necessary constructors, accessors and modifiers for the *Student* class.
4. Change the datatype of *data* attribute of *Node* class to *Student*.
5. Update the constructors, accessors and modifiers of *Node* class where needed.
6. Implement the following menus in your main program:

Choose an option:

I - Insert

D - Display

S – Stop

where **Insert** is used to insert a student detail to the beginning of the list, **Display** is to show all the data from the list in a tabular format and **Stop** is to terminate the program.

7. Test your program by inserting and displaying three student details.

Question 3 – Enhancing the solution

To improve the solution to Question 2, make the following changes:

1. Create a function void `assign_grade()` should be the behaviour of the *Student* class. Make other required changes so your program functions properly.
2. Create a text file with student information, then read the information from the text file rather than the keyboard.